# Feature Speed, Transaction Costs, and the Limits of Machine Learning in Emerging Markets

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## Abstract

Across 15 emerging markets, eight portfolio interventions converge on the same net Sharpe ceiling (~0.45–0.49 at 3× transaction costs), regardless of whether they restrict features, smooth predictions, or reduce rebalancing frequency. We show this ceiling is traced by a cost-speed frontier: $\mathrm{SR}_{\mathrm{gross}} - \mathrm{SR}_{\mathrm{net}} \approx \lambda\kappa \cdot \mathrm{FSI} + c$, where the Feature Speed Index (FSI) measures a strategy's ex-ante exposure to fast-moving characteristics. FSI explains 64–65% of cross-sectional SR degradation at each cost level ($\lambda = 0.33$ per unit $\kappa$; pooled $R^2 = 0.72$ across 300 strategy–market–cost combinations) and predicts realized turnover ex ante ($R^2 = 0.73$). Non-smooth models (trees, neural networks) fall systematically below the frontier, revealing model smoothness as a second constraint. The frontier explains why ML's prediction advantage (gross SR 0.68 vs. sorts' 0.58) vanishes after costs: ML loads on fast features that generate 2–3× the turnover of sorts. Restricting Ridge regression to slow, persistent characteristics is the simplest path to the ceiling (SR 0.47, 10% turnover); EMA smoothing and quarterly rebalancing reach the same frontier through different mechanisms, while cost-aware neural networks trade away signal for turnover and fall short.

## CCS Concepts

• **Mathematics of computing** → **Probabilistic algorithms**; • **Computing methodologies** → **Machine learning**.

## Keywords

machine learning, emerging markets, transaction costs, portfolio construction, factor investing

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## 1 Introduction

Machine learning has transformed cross-sectional asset pricing. Gu, Kelly, and Xiu [8] show that tree-based and neural network models produce superior out-of-sample $R^2$ for US equities, and Hanauer and Kalsbach [9] extend this to emerging markets (EM). More recently, Jensen, Kelly, Malamud, and Pedersen [10] (JKMP) identify that cost-agnostic ML overweights "fleeting small-scale characteristics," proposing an "implementable efficient frontier" that integrates costs into the optimization objective.

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We build on their diagnosis and show that the cost–speed interaction creates a *performance ceiling*: across 15 emerging markets, eight intervention variants spanning four mechanism families (feature selection, persistence filtering, prediction smoothing, quarterly rebalancing) all converge on net SR ~0.45–0.49 at 3× EM costs. Where JKMP optimize toward an implementable frontier, we characterize the ceiling that any method (upstream or downstream) approaches, and provide an ex-ante diagnostic, the Feature Speed Index (FSI), computable from training data alone. Our central contribution is the frontier as an ex-ante diagnostic: FSI, computed from training data alone, explains 64–65% of cross-sectional cost-driven SR degradation at every cost level and 73% of realized turnover.

Five findings anchor the analysis, using 154 firm-level characteristics from the JKP Global Factor Data [11], walk-forward validation, and Corwin-Schultz [3] transaction costs:

(1) **A cost-speed ceiling.** Eight intervention variants converge on net SR ~0.45–0.49 despite different mechanisms. The frontier $\mathrm{SR}_{\mathrm{gross}} - \mathrm{SR}_{\mathrm{net}} \approx \lambda\kappa \cdot \mathrm{FSI} + c$ fits SR degradation with pooled $R^2 = 0.72$ across 300 strategy–market–cost combinations; at each fixed cost level, FSI explains 64–65% of cross-sectional degradation ($\lambda = 0.33$ per unit $\kappa$).
(2) **Cost–speed interaction.** At zero costs, fast and slow features produce comparable gross SRs (0.63 vs. 0.61). At 3× EM costs, fast features degrade ~3× faster (SR degradation 0.42 vs. 0.14).
(3) **Simple upstream fixes reach the ceiling.** Restricting Ridge to slow features (57 characteristics) achieves net SR 0.47 at 10% turnover vs. standard Ridge (0.35, 23%) and MomentumSort (0.36, 17%), outperforming the sort in 11/15 markets (not individually significant; Section 6).
(4) **ML overweights fast features.** ML allocates 43–47% of importance to fast features (25% of the set), driving 63% of prediction changes.
(5) **Prediction stability is the binding constraint.** Tree models pair the strongest gross signals with the worst net SRs (XGBoost 0.66 gross, 0.17 net; Table 1) and slow features do not help; walk-forward EMA smoothing lifts XGBoost only to sort parity (0.35), not to the frontier. Prediction stability, not accuracy, is the binding constraint.

A cross-regime validation on US equities (top 500, 2021–2025) confirms the decomposition: Ridge_Slow outperforms Ridge_Fast at every cost level, and Ridge_Fast collapses at 3× costs (SR −0.19 vs. 0.51). Our results complement JKMP's downstream approach with a simpler upstream alternative, and provide large-scale EM evidence for Gârleanu and Pedersen's [7] prediction that optimal portfolios overweight slow-decaying signals. The EM setting, with 3–5× higher costs, amplifies the interaction into a quantifiable ceiling.

## 2 Related Work

*Cost-aware ML..* JKMP [10] integrate transaction costs into the ML portfolio optimization objective, identifying that cost-agnostic ML overweights "fleeting" characteristics. We differ in three respects: (i) a quantitative feature-level diagnostic (rank autocorrelation classifies which features drive turnover), (ii) demonstration that the problem is cost-regime dependent (modest in US, catastrophic in EM), and (iii) a simpler upstream fix (slow-feature selection, net SR 0.47 vs. MomSort 0.36) that avoids cost-aware optimization entirely.

*ML for asset pricing.* Gu et al. [8] establish ML's superiority for US equity prediction. Hanauer and Kalsbach [9] extend this to EM, finding ML generates higher long-short Sharpe ratios than linear baselines. We confirm the ML-vs-linear ranking but show that after costs in long-only portfolios, sorts match or beat ML. Esakia and Goltz [5] reach a similar conclusion for long-only settings. Azevedo et al. [1] find 120–140% monthly turnover and 13–40% performance reduction from costs in ML strategies. Chen et al. [2] document that researcher degrees of freedom produce non-standard errors ∼1.59× the standard error.

*Optimal trading with slow signals.* Gârleanu and Pedersen [7] (GP) prove that under transaction costs, optimal portfolios overweight slow-decaying signals. Our results validate GP's predictions in a setting they did not study: ML portfolios with 154 characteristics across 15 EM markets.

*Transaction costs and implementability.* Realistic trading costs render many anomalies unprofitable [13], favor diversified factor combinations [4], and reward low-turnover construction [6], including turnover-aware EM factor designs [14].

## 3 Data and Methodology

### 3.1 Data

We use monthly equity data for 15 emerging markets from the JKP Global Factor Data [11]: BRA, CHN, GRC, IDN, IND, KOR, MEX, MYS, PHL, POL, SAU, THA, TUR, TWN, and ZAF. Each provides 154 firm-level characteristics. The prediction target is the one-month excess return (`ret_1_0`). For each market-month, we restrict to the top $N$ stocks by market cap ($N \in \{50, 300\}$; most at 100). Features are lagged *before* universe filtering to avoid collider bias, since $\text{me}(T) = \text{me}(T-1) \times (1+r_T)$.

### 3.2 Strategies

We compare seven strategies: three factor sorts (MomentumSort: 12-month return with 1-month skip; LowVolSort: 60-day volatility; CompositeSort: $z$-score of momentum, value, and quality), three ML models (Ridge $\alpha$=100; XGBoost depth 4, 200 trees; LightGBM 31 leaves, 200 trees), and a MomentumML_Blend (80% momentum + 20% Ridge rank). All use top-quintile, long-only, equal-weighted portfolios with a 20% rebalancing buffer. All ML models train on cross-sectional rank targets: $\tilde{y}_{i,t} = \text{Rank}(r_{i,t})/N_t$.

### 3.3 Walk-Forward Validation

We use expanding-window walk-forward with 60-month minimum training. Models retrain monthly on all available data up to time $t$. The OOS period begins 2016–2018 (market-dependent), yielding 55–93 OOS months.

### 3.4 Transaction Cost Model

Costs use the Corwin-Schultz [3] half-spread with multiplier $\kappa$: $c_{i,t} = \kappa \times \text{HalfSpread}_{i,t}$, clipped to $[20, 200]$ bps. Our baseline is $\kappa = 3$, following Novy-Marx and Velikov [13] and validated against actual costs from a mid-size EM equity fund (∼70–100 bps one-way). Median CS half-spreads on our EM universes are 30–40 bps versus ∼25 bps for US large-caps; the EM cost gap is driven mainly by market impact from thinner order books, captured by $\kappa$. For robustness, we also test flat cost schedules (10–100 bps per trade).

### 3.5 Performance Metrics

Sharpe ratios are computed as $\text{SR} = (\bar{r}_p/\sigma_p) \times \sqrt{12}$, where $r_p$ is the monthly net-of-cost portfolio return. Since all strategies are long-only in the same cap-filtered universe, relative comparisons are unaffected by the common market-beta component. We additionally report an *alpha Sharpe ratio* $\text{SR}_\alpha = \text{SR}(r_p - r_{\text{EW}})$ that isolates stock-selection skill by subtracting the equal-weight universe return each month.

### 3.6 The Feature Speed Index and the Cost-Speed Frontier

To quantify ex-ante exposure to prediction instability, we define the Feature Speed Index (FSI) as the equal-weight average instability of a strategy's feature set $S$:

$$\text{FSI}(S) = \frac{1}{|S|} \sum_{j \in S} (1 - \text{AC}_j), \tag{1}$$

where $\text{AC}_j$ is the cross-sectional Spearman rank autocorrelation of feature $j$, computed on a rolling 60-month training window. This feature-set definition is available before model fitting for all strategies and therefore avoids using realized model coefficients in the diagnostic. For single-feature sorts, $S$ contains only the sorting characteristic.

The FSI enters a linear frontier for cost-driven performance loss:

$$\text{SR}_{\text{gross}} - \text{SR}_{\text{net}} \approx \lambda\kappa \cdot \text{FSI} + c, \tag{2}$$

where $\kappa$ is the cost multiplier, $\lambda$ captures the sensitivity of degradation to feature speed, and $c$ is the baseline cost drag. Under proportional costs, $\kappa$-scaling is mechanical; the frontier's informative content is *cross-sectional* (Section 4.6). It is estimated on smooth models (Ridge, sorts) where prediction surfaces are continuous; non-smooth models exhibit systematic excess degradation beyond it (Section 4.5).

## 4 Results

### 4.1 The Cost–Speed Interaction

We classify 154 features into three speed categories: **SLOW** (57 accounting fundamentals, rank AC 0.97), **MEDIUM** (56 intermediate features, AC 0.83), and **FAST** (38 market microstructure variables, AC 0.68). The classification is validated empirically: the three classes have significantly different rank AC (Kruskal-Wallis $p = 3.77 \times 10^{-11}$).

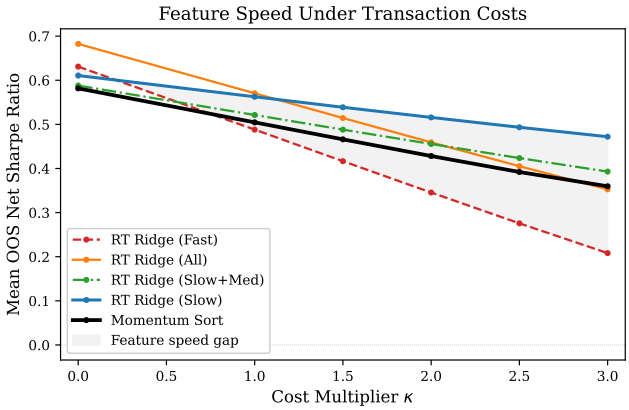



**Figure 1: Net SR vs. cost multiplier $\kappa$. Fast features degrade ∼3× faster than slow features. The shaded region shows the feature-speed gap, which widens monotonically with $\kappa$. At zero costs, fast and slow are competitive; at 3× EM costs, the gap exceeds 0.25 SR.**

**Table 1: Gross vs. net Sharpe ratios at 3× costs, averaged across 15 EM markets. Ridge has a stronger gross signal but higher turnover erases the advantage.**

| Strategy | Gross SR | Net SR | Turnover | Cost Drag |
|---|---|---|---|---|
| MomentumSort | 0.58 | 0.36 | 17.0% | ∼590 bps |
| Ridge | 0.68 | 0.35 | 23.3% | ∼700 bps |
| XGBoost | 0.66 | 0.17 | 36.3% | ∼1,100 bps |
| LightGBM | 0.64 | 0.13 | 38.2% | ∼1,160 bps |
| Blend (80/20) | 0.63 | 0.38 | 19.0% | ∼640 bps |

Figure 1 plots net Sharpe ratio against the cost multiplier $\kappa$. At zero costs, fast features produce competitive gross SRs (0.63 vs. 0.61 for slow); the predictive content of fast features is not in question. As costs rise, fast-feature performance degrades ∼3× faster (degradation 0.42 vs. 0.14 SR at 3×), because turnover differences compound linearly with $\kappa$. This is the central result: *the feature-speed problem is not a signal failure but a cost interaction.*

In EM, the cost–speed interaction is 3–5× larger than in US markets, suggesting that downstream optimization alone may not suffice; the problem must also be addressed upstream.

## 4.2 ML vs. Sorts: Ridge Matches Momentum

With rank-target training, Ridge essentially matches MomentumSort across 15 EM markets at 3× costs: average net SRs are Blend 0.38, MomSort 0.36, Ridge 0.35, CompositeSort 0.32, XGBoost 0.17, LightGBM 0.13. Ridge wins 8/15 markets (Wilcoxon $p = 0.45$, not significant); tree models are significantly worse ($p < 0.01$).

Table 1 isolates the cost interaction. Ridge generates a *stronger* gross signal (0.68) than MomSort (0.58), but 23% turnover vs. 17% erases the advantage.

Per-market results (Table 2) show Ridge wins 8/15 markets, with advantages where turnover is contained (TWN: 0.73 vs. 0.51; CHN: 0.23 vs. 0.07). Ridge_Slow (Section 4.4) is best in 5 markets, positive in 13/15.

## 4.3 The Feature-Speed Decomposition

The cost–speed interaction (Section 4.1) establishes that fast features degrade faster under costs. We now decompose *why* ML preferentially loads on these features and trace the chain from feature characteristics to portfolio turnover.

Table 3 reports three diagnostics. FAST features represent 25% of the feature set but receive 43–47% of model importance and generate 63% of month-to-month prediction changes. Their low rank autocorrelation (0.68 vs. 0.97 for SLOW) makes predictions inherit their instability.

*Feature overweighting.* FAST features (25% of the set) receive 43% of importance in Ridge and 47% in XGBoost, drawn by high-frequency signal-to-noise advantages in the training objective.

*Stock-level IC confirms the tradeoff.* Table 4 quantifies the prediction-stability tradeoff. Ridge_Slow achieves the highest IC×stability (0.030), pairing a momentum-level IC (0.031) with near-perfect stability (0.939); Ridge_Fast has the lowest stability (0.539) with no IC compensation, confirming fast features as the primary instability source.

*Model-free signal decay.* The speed differential is a data property, not a modeling artifact. With no fitted model, we compute per-feature monthly rank ICs against single-month returns at horizons of 1–12 months, sign-aligned per feature. One-month ICs are comparable (slow 0.027, fast 0.025), but slow features retain 69% of their IC at a 12-month horizon versus 57% for fast features (Wilcoxon $p = 0.01$ across 15 markets; Figure 2). Notably, fast features' predictive *content* decays only moderately faster; it is their *values* that churn (rank AC 0.68 vs. 0.97, Table 3), which is what forces turnover. This confirms the cost interaction of Section 4.1: fast signals remain informative, but trading them is expensive.

*Summary.* FAST features $\xrightarrow{\text{high importance}}$ unstable predictions $\xrightarrow{\text{high turnover}}$ low net SR. This structural tension between predictive power and tradability produces a modest drag in low-cost environments and strategy failure in high-cost EM.

## 4.4 Upstream Fix: Slow-Feature Ridge

The decomposition motivates a simple upstream fix: restrict features to slow, persistent characteristics before training. Table 5 confirms this dramatically. Ridge_Slow achieves net SR 0.47 at 10% turnover versus MomSort's 0.36 at 17%, with the *same model and hyperparameters*. The improvement comes entirely from removing fast features. (A pooled block bootstrap yields $p = 0.10$ for the SR difference; see Section 6. Ridge_Slow outperforms in 11/15 markets and 90% of bootstrap samples; detecting an effect of this magnitude at 5% would require ∼80 years of OOS data.)

*Aggregation versus speed.* Comparing Ridge and CompositeSort at matched dimensionality ($N \in \{5, 10, 20, 50\}$): at $N = 5$, turnover is nearly identical (5.1% vs. 5.6%), ruling out regression itself as the turnover source.

**Table 2: Out-of-sample net Sharpe ratios at 3× costs across 15 emerging markets. $N$ is the universe size (top stocks by market cap); $T_{OOS}$ is the number of out-of-sample months. Bold indicates the best active strategy per market. EW is the equal-weighted passive benchmark. Ridge matches MomSort on average (0.35 vs. 0.36) but generates higher turnover (23% vs 17%); Ridge_Slow (Section 4.4) reaches 0.47 at 10% turnover.**

| Market | $N$ | $T_{OOS}$ | EW | MomSort | Composite | Ridge | Ridge_Slow | XGBoost | LightGBM | Blend | $TO_{Mom}$ | $TO_{Ridge}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| BRA | 100 | 83 | 0.10 | **0.08** | −0.04 | −0.05 | −0.06 | −0.30 | −0.23 | **0.08** | 18% | 30% |
| CHN | 300 | 78 | 0.24 | 0.07 | 0.13 | 0.23 | **0.39** | −0.07 | −0.17 | 0.08 | 21% | 23% |
| GRC | 50 | 78 | 0.64 | 0.43 | **0.92** | 0.73 | 0.74 | 0.64 | 0.50 | 0.49 | 15% | 19% |
| IDN | 100 | 80 | 0.29 | **0.59** | 0.13 | 0.42 | 0.54 | 0.22 | 0.14 | 0.47 | 17% | 18% |
| IND | 200 | 81 | 0.61 | **0.63** | 0.29 | 0.37 | 0.58 | 0.09 | −0.11 | 0.59 | 16% | 28% |
| KOR | 200 | 86 | 0.07 | 0.03 | −0.32 | **0.07** | 0.07 | −0.23 | −0.29 | −0.04 | 18% | 15% |
| MEX | 100 | 93 | 0.46 | 0.25 | 0.40 | 0.18 | **0.51** | −0.24 | −0.32 | 0.23 | 18% | 27% |
| MYS | 100 | 90 | 0.23 | 0.39 | 0.14 | **0.46** | 0.43 | 0.24 | 0.28 | **0.46** | 15% | 22% |
| PHL | 100 | 87 | −0.02 | 0.01 | **0.41** | 0.10 | 0.16 | −0.14 | −0.11 | 0.17 | 15% | 25% |
| POL | 100 | 69 | 0.53 | 0.55 | 0.60 | 0.57 | **0.70** | 0.60 | 0.57 | 0.55 | 17% | 24% |
| SAU | 100 | 56 | 1.01 | 1.13 | 0.93 | 0.85 | 1.20 | 0.93 | 0.94 | **1.22** | 14% | 25% |
| THA | 100 | 84 | −0.09 | 0.01 | −0.18 | −0.05 | −0.06 | −0.38 | −0.43 | **0.09** | 17% | 20% |
| TUR | 100 | 79 | 0.61 | 0.33 | **0.67** | 0.45 | 0.54 | 0.39 | 0.43 | 0.35 | 18% | 24% |
| TWN | 200 | 78 | 0.67 | 0.51 | 0.54 | 0.73 | **0.86** | 0.71 | 0.73 | 0.51 | 17% | 23% |
| ZAF | 100 | 93 | 0.30 | 0.39 | 0.24 | 0.24 | **0.49** | 0.07 | −0.02 | 0.48 | 18% | 28% |
| **Mean** | – | 80 | 0.38 | 0.36 | 0.32 | 0.35 | **0.47** | 0.17 | 0.13 | 0.38 | 17% | 23% |

**Table 3: Feature speed decomposition. FAST features dominate both model importance and prediction changes despite representing a minority of the feature set.**

| Speed | # Feat. | Rank AC | % Import. | % Pred. Δ |
|---|---|---|---|---|
| SLOW | 57 | 0.97 | 31–34% | 12.6% |
| MEDIUM | 56 | 0.83 | 22–23% | 24.1% |
| FAST | 38 | 0.68 | 43–47% | 63.4% |

**Table 4: Information coefficient analysis across 15 EM markets. IC×Stability captures the effective signal after accounting for prediction churn.**

| Strategy | IC | Stability | IC×Stab | Hit Rate |
|---|---|---|---|---|
| MomentumSort | 0.031 | 0.894 | 0.028 | 57.6% |
| Ridge_All | 0.022 | 0.665 | 0.015 | 57.0% |
| Ridge_Slow | 0.031 | 0.939 | **0.030** | 57.7% |
| Ridge_Fast | 0.013 | 0.539 | 0.007 | 54.0% |

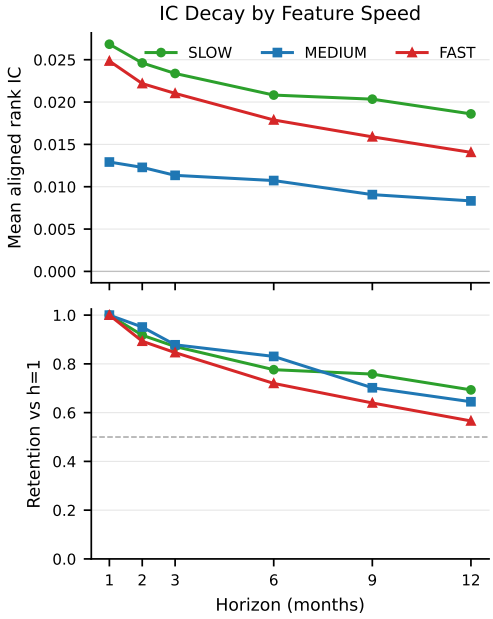



**Figure 2: Model-free rank-IC decay by feature speed class. Slow features retain 69% of their one-month IC at 12 months versus 57% for fast features; no model is fitted anywhere in this analysis.**

*Data-driven persistence filter.* A persistence filter (rank AC > 0.75, recomputed on training data; ∼120 features) and a speed-tercile split (∼127 features) both yield net SR 0.47 at 13% turnover, matching the ad-hoc result.

This upstream fix requires no cost-aware optimization, no prediction smoothing, and no change to portfolio construction. It is the simplest intervention with the largest improvement.

*Alpha over the passive benchmark.* To isolate stock-selection skill, we compute $SR_\alpha = SR(r_p - r_{EW})$, where $r_{EW}$ is the equal-weighted universe return. At 3× costs: Ridge_Slow $SR_\alpha$ = +0.18 (positive in 10/15 markets), MomSort +0.07 (8/15), Ridge_All −0.15 (6/15),

**Table 5: Feature-speed ablation: Ridge with different feature subsets. Mean OOS net SRs across 15 markets at 3× costs.**

| Strategy | Net SR | Turnover | # Features |
|---|---|---|---|
| **Ridge_Slow** | **0.47** | **10.2%** | 57 |
| Ridge_SlowMed | 0.39 | 14.5% | 113 |
| MomentumSort | 0.36 | 17.0% | 1 |
| Ridge_All | 0.35 | 23.3% | 151[†] |
| Ridge_Fast | 0.21 | 30.1% | 38 |

[†]154 characteristics minus me (universe filter), target, and identifier.

**Table 6: The prediction-portfolio disconnect across model classes and smoothing interventions. Sorted by gross SR. All rows use the tree-comparison sample: walk-forward over all months after the initial 60-month training window (~102 months per market), a longer window than Tables 1–2's OOS period, so levels are not comparable across tables.**

| Model | Strategy | Gross SR | Net SR | SR Lost | TO |
|---|---|---|---|---|---|
| XGB | All | 0.767 | 0.300 | −0.467 | 34.5% |
| LGBM | All | 0.748 | 0.265 | −0.482 | 35.5% |
| Ridge | All | 0.684 | 0.357 | −0.326 | 23.0% |
| XGB | Slow | 0.647 | 0.263 | −0.385 | 28.6% |
| Ridge | Slow | 0.616 | 0.479 | −0.137 | 9.9% |
| MomSort | — | 0.606 | 0.373 | −0.233 | 17.2% |

Ridge_Fast −0.46 (2/15). At 1× costs, all variants produce comparable positive alpha, confirming the gap is entirely cost-mediated.

## 4.5 Model Complexity Amplifies the Cost Channel

We extend the analysis to tree-based models, revealing a second mechanism, model complexity, that amplifies the disconnect independently of feature speed.

Table 6 presents gross-to-net performance across all model–feature combinations on a common sample. XGBoost achieves the highest unsmoothed gross SR (0.77), yet its net SR (0.30) falls well below Ridge_All (0.36) and Ridge_Slow (0.48). The gross ranking is strongly inverted in net terms (rank correlation −0.52 across model–feature combinations).

*Slow features do not help trees; smoothing only partly does.* XGB_Slow (net SR 0.263) performs *worse* than XGB_All (0.300), because decision boundary shifts generate turnover independent of input stability. Output smoothing is the more effective lever, but only partially: on the main OOS sample, walk-forward EMA lifts XGBoost's net SR from 0.24 to 0.35 (turnover 36% to 25%), recovering sort parity (0.36) but not the frontier. Even the best fixed $\alpha = 0.2$, selected post hoc, reaches 0.45 at 12% turnover, approaching but not exceeding Ridge_Slow's 0.47. A post-2020 validation had suggested smoothed trees exceed Ridge_Slow (0.53); this does not replicate on the full sample.

*A 2 × 2 framework.* The results decompose into feature speed × model smoothness (Figure 3): Ridge_Slow loses only 0.14 SR to costs, Ridge_All 0.33, XGB_Slow 0.39, XGB_All 0.47. Output smoothing

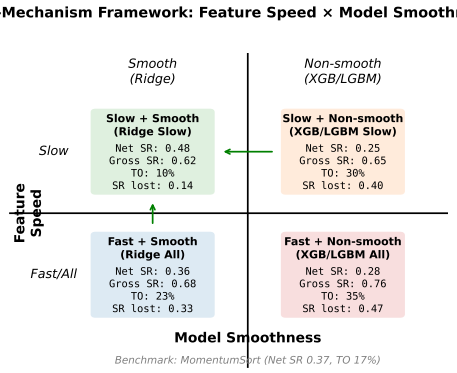



**Figure 3: The 2×2 framework: feature speed × model smoothness. Both slow features and a smooth model are needed to minimize the prediction-portfolio disconnect.**

partially bridges the gap (walk-forward EMA cuts XGBoost's gross-to-net loss from 0.49 to 0.33 on the main OOS sample), confirming that prediction instability, not model capacity, is the mechanism.

*Implication.* The binding constraint is prediction *stability*, not accuracy. EMA is a universal fix: it improves Ridge (+0.10 SR), transforms XGBoost (+0.44 SR), but cannot fully compensate for piecewise-constant surfaces. Trees' superior gross signal (0.91 with EMA vs. 0.63 for Ridge_Slow) is partially offset by residual instability, explaining why simple linear models remain competitive after costs [5, 8].

## 4.6 The Cost-Speed Frontier

We estimate the frontier (Equation 2) on smooth models (Ridge variants, sorts) across 300 strategy–market–cost combinations (5 strategies × 15 markets × 4 cost levels). Figure 4 plots $\kappa \cdot \text{FSI}$ against SR degradation: pooled $R^2 = 0.72$. Because costs are proportional to $\kappa$, per-$\kappa$ slopes of $0.33\kappa$ are arithmetic, not evidence; they confirm only that these strategies trade identically at every cost level. The substantive result is cross-sectional: at each fixed $\kappa$, FSI alone explains 64–65% of degradation across the 75 strategy–market cells ($\kappa$ alone: 37% pooled). We quote the per-$\kappa$ slope $\lambda = 0.33$ throughout (the pooled slope, 0.39, overweights high-$\kappa$ cells; cluster-robust 95% CIs: $[0.27, 0.40]$ by market, $[0.26, 0.41]$ by strategy). FSI is equally an ex-ante turnover diagnostic (it explains 73% of realized turnover across cells) and is computable from training data *before* any trading occurs. Its resolution is at the strategy level: FSI is a feature-set property, nearly constant across markets within a strategy, so it separates feature sets rather than markets (within-strategy $R^2 = 0.07$); its coefficient is unchanged when controlling for each market's spread level.

Pooling tree models (XGBoost, LightGBM; same OOS window as Table 1) with a non-smoothness indicator turns the second channel into a joint test rather than a residual description: at $\kappa = 3$, $R^2$ rises from 0.34 to 0.73 with +0.22 SR excess tree degradation and an unchanged FSI coefficient. EMA output smoothing moves trees toward the frontier without reaching it (walk-forward: 0.24 to 0.35; best post-hoc fixed $\alpha$: 0.45 at 12% TO).

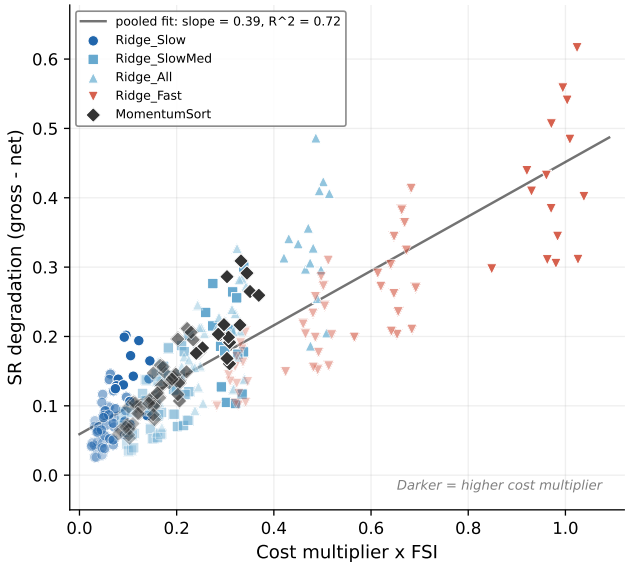



**Figure 4: The cost-speed frontier (Equation 2): SR degradation vs. $\kappa \times$ FSI across 300 strategy–market–cost combinations. Pooled $R^2 = 0.72$; cross-sectional $R^2 = 0.64$–$0.65$ at each fixed $\kappa$ with slope $\lambda = 0.33$ per unit $\kappa$. Darker points indicate higher $\kappa$.**

Figure 5 visualizes the resulting frontier. At 3× EM costs, eight strategies cluster near net SR 0.45–0.49 despite radically different mechanisms: slow-feature selection (Ridge_Slow 0.47), persistence filtering (0.47), speed-tercile selection (0.47), a data-driven rank-target variant (0.46), fixed and walk-forward EMA smoothing (0.47 and 0.45), and quarterly rebalancing (Ridge_All_Q 0.48, Ridge_Slow_Q 0.49). Strategies that fail to minimize FSI (tree models, fast-feature Ridge, unconstrained NNs) fall below the ceiling in proportion to $\kappa \cdot$ FSI.

The ceiling height depends on $\text{SR}_{\text{gross}}$ and the market's cost environment; the frontier's portable content is the cross-sectional FSI–degradation relationship at each cost level.

*Cross-market transfer.* To test whether the frontier generalizes out of sample, we run leave-one-market-out (LOMO) and leave-5-out cross-validation. LOMO (15 folds) yields mean OOS $R^2 = 0.60$; leave-5-out (3,003 splits) yields mean OOS $R^2 = 0.69$, with 97% of splits above 0.50. The fitted slope is stable across folds (std = 0.008, range $[0.37, 0.40]$; bootstrap 95% CI $[0.34, 0.45]$), so the cross-sectional relationship is not driven by any single market. The slow-feature set also transfers: 52 of 57 slow features are classified as slow in *all* 15 markets (mean Jaccard similarity 0.97), so "slow" can be defined on one market subset and applied elsewhere.

## 4.7 Prediction Smoothing

A complementary intervention smooths predictions at the output level. We apply an EMA to Ridge's prediction ranks: $\hat{r}_{i,t}^{\text{smooth}} = \alpha \cdot \hat{r}_{i,t}^{\text{raw}} + (1-\alpha) \cdot \hat{r}_{i,t-1}^{\text{smooth}}$, where lower $\alpha$ means more smoothing. Figure 6 shows the alpha–SR relationship. The curve peaks at $\alpha = 0.35$ (net SR 0.47), forming an inverted-U. Walk-forward alpha selection, maximizing IS net SR over the last 24 training months scored by the current-window model, achieves 0.45 without using future OOS months, beating MomSort (0.36) in 10/15 markets.

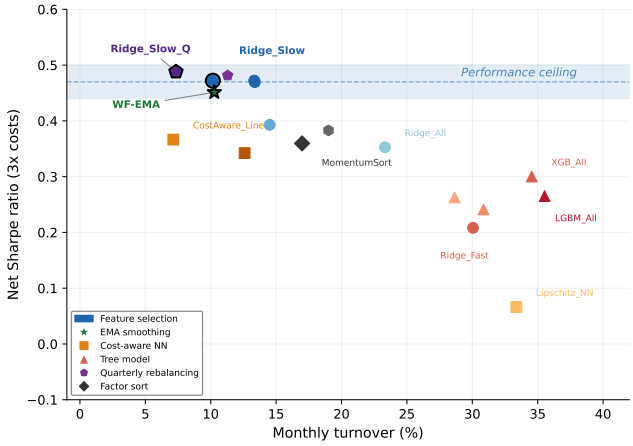



**Figure 5: The cost-speed frontier at 3× EM costs. Strategies that minimize FSI converge on net SR ~0.45–0.49 (shaded band). Below the ceiling, net SR degrades with turnover along the frontier. Color indicates mechanism type.**

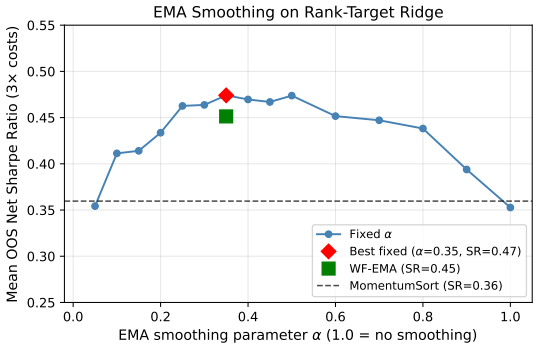



**Figure 6: EMA smoothing: $\alpha$ vs. net SR at 3× costs (15 markets). Walk-forward EMA (green square) achieves 0.45, beating MomSort (0.36).**

Table 7 quantifies the gains. WF-EMA's gross SR (0.60) is *below* unsmoothed Ridge (0.68): the +0.10 net SR improvement comes entirely from turnover reduction (23% → 10%). EMA helps Ridge but *hurts* MomSort (−0.04), confirming it recovers ML-specific signal.

*EMA is redundant with slow features.* Ridge_Slow with EMA selects $\alpha = 1.0$ (no smoothing): when features are already slow, predictions are inherently stable (rank stability 0.939, Table 4). The two interventions are substitutes.

*EMA helps tree models, but less reliably.* EMA is not Ridge-specific: it lifts XGBoost from 0.24 to 0.35 (Section 4.5). Two differences from Ridge: the optimal smoothing is more aggressive ($\alpha = 0.2$ vs. 0.35), reflecting trees' greater inherent instability, and walk-forward

**Table 7: Prediction smoothing results (15 markets). WF-EMA selects $\alpha$ inside the walk-forward loop using recent in-sample months and does not use future OOS months.**

| Strategy | Net SR | Gross SR | TO | Stability | Wins |
|---|---|---|---|---|---|
| Ridge (no smooth) | 0.35 | 0.68 | 23.3% | 0.750 | 8/15 |
| MomentumSort | 0.36 | 0.58 | 17.0% | — | — |
| **Ridge WF-EMA** | **0.45** | **0.60** | **10.3%** | **0.919** | **10/15** |

**Table 8: US equities: net SR by strategy and cost multiplier. Top 500, OOS 2021–2025. Ridge_Slow outperforms Ridge_Fast at every cost level; Ridge_Fast goes negative at 3×.**

| $\kappa$ | MomSort | Ridge_All | Ridge_Slow | Ridge_Fast |
|---|---|---|---|---|
| 0× | 0.775 | 0.580 | 0.638 | 0.404 |
| 1× | 0.693 | 0.445 | 0.574 | 0.206 |
| 2× | 0.612 | 0.310 | 0.510 | 0.008 |
| 3× | 0.530 | 0.175 | 0.510 | −0.189 |

selection captures barely half the post-hoc gain (+0.11 of +0.21; Ridge captures +0.10 of +0.12), because trees' noisier in-sample net SR misguides the $\alpha$ choice.

## 4.8 Cross-Regime Validation: US Equities

We test the identical pipeline on US equities (top 500 by market cap, OOS 2021–2025, 50 months). Table 8 confirms the decomposition is not EM-specific: Ridge_Slow outperforms Ridge_Fast at every cost level, and Ridge_Fast collapses at 3× costs (−0.19 vs. 0.51). MomentumSort retains a clear advantage in the US; we do not claim Ridge beats MomSort there. The EM–US divergence reflects two channels: MomSort's gross signal is substantially stronger in the US (0.78 vs. 0.58 in EM), and EM costs are 3–5× larger, making the turnover reduction from slow-feature selection more valuable relative to the sort's advantage. The key point is that the *relative* patterns within Ridge's feature subsets replicate precisely.

*Two channels.* The decomposition operates through signal degradation and cost amplification. Ridge_Fast trails Ridge_Slow even at zero costs (0.40 vs. 0.64), and the gap widens from 0.23 to 0.70 SR as $\kappa$ rises from 0 to 3. In EM, ML concentrates importance on fast features well beyond their share of the feature set (Table 3), compounding both channels.

# 5 Discussion

*Theoretical foundation: GP, JKMP, and the ceiling.* Our cost-speed frontier (Equation 2) operationalizes Gârleanu and Pedersen's [7] (GP) insight that optimal portfolios overweight slow-decaying signals. GP prove this in continuous time with known dynamics; we show the same structure creates a quantifiable ceiling in a discrete-time ML pipeline with 154 estimated characteristics. A per-$\kappa$ slope of exactly $\lambda\kappa$ under proportional costs means strategies trade identically at every cost level. This is a diagnosis, not a discovery: cost-aware strategies (JKMP-style) would cut turnover as $\kappa$ rises, bending the frontier ($\lambda$ decreasing in $\kappa$). Standard ML pipelines show no such adaptation, motivating the upstream fix. JKMP [10] optimize downstream (cost-aware training); we characterize the ceiling that any method, upstream or downstream, approaches. In EM, where cost estimates are noisy, the upstream fix reaches the ceiling without requiring reliable cost inputs.

The $2 \times 2$ framework (Section 4.5) reveals why non-smooth models cannot reach the ceiling: model smoothness is a second constraint that feature selection alone cannot address. On the same slow features, trees and unconstrained NNs exhibit excess degradation correlated $r = 0.91$ with model-induced turnover. Both slow inputs *and* smooth models are needed.

*Reconciling with Hanauer and Kalsbach.* Hanauer and Kalsbach [9] show ML produces superior long-short Sharpe ratios vs. linear models in EM. We confirm their core finding: ML generates stronger *gross* signals (Ridge gross SR 0.68 vs. MomSort 0.58). The divergence arises from the evaluation framework: we benchmark against factor sorts (sparse, low-turnover, and thus a harder cost-adjusted benchmark); we evaluate long-only portfolios, since short-selling is restricted in most EM (Section 6); and at realistic EM costs ($\kappa = 3$, ∼90 bps one-way), ML's turnover from fast-feature loading erases the signal advantage. Our results complement rather than contradict Hanauer: they identify the implementation boundary beyond which ML's signal superiority no longer translates to portfolio outperformance.

*Practical recommendations.* For EM equity portfolios: (1) without ML infrastructure, default to momentum sorts; (2) teams with ML should restrict to slow features and train on rank targets, which beats the sort without prediction smoothing or cost-aware optimization. In our data, fast features' gross advantage survives only below $\kappa \approx 0.2$ (∼6 bps one-way): at all realistic EM cost levels, slow features dominate. The FSI provides an ex-ante screen: compute Equation 1 before deployment, and if FSI exceeds ∼0.15, restrict features or apply smoothing.

*Limitations.* Our OOS spans 55–93 months per market with cross-country correlation ∼0.6, limiting statistical power (see Section 6 for formal tests). The Corwin-Schultz cost model with 3× multiplier is validated against an EM equity fund but remains a proxy. We test only large-caps; ML may fare better in small-caps, but costs there are higher. Tree hyperparameters are fixed at standard values [8], making our comparison conservative. Implementation choices affect results [2], though the feature-speed decomposition reflects data properties rather than modeling choices. Combining upstream feature selection with JKMP-style downstream optimization is untested.

# 6 Robustness and Alternatives

*Cost-aware neural networks.* Cost-aware training objectives could address turnover without discarding features. We test three alternatives, *all restricted to the 57 slow features*: (i) **CostAware_Linear** (linear model with differentiable net-of-cost Sharpe loss), (ii) **CostAware_Lipschitz** (spectral-norm MLP with cost penalty), and (iii) **Lipschitz_NN** (spectral-norm MLP without cost penalty), all with rank targets and walk-forward HP selection. Ridge_Slow remains best (Table 9). CostAware_Linear achieves 7.1% turnover but sacrifices gross signal (net SR 0.37). Lipschitz_NN fails entirely

**Table 9: Cost-aware alternatives vs. Ridge_Slow (rank targets). Mean OOS across 15 markets at 3× costs. No cost-aware variant matches the upstream fix.**

| Strategy | Net SR | Turnover | Gross SR |
|---|---|---|---|
| **Ridge_Slow** | **0.47** | 10.2% | 0.61 |
| CostAware_Linear | 0.37 | 7.1% | 0.47 |
| MomentumSort | 0.36 | 17.0% | 0.58 |
| CostAware_Lipschitz | 0.34 | 12.6% | 0.52 |
| Lipschitz_NN | 0.07 | 33.4% | 0.48 |

(SR 0.07, 33% turnover): even with slow inputs, the nonlinear prediction surface generates instability. Coupling signal and turnover in a single loss forces a tradeoff that the simple upstream fix avoids.

*Long-short portfolios.* Short-selling is restricted or prohibitively expensive in most EM. On five markets (BRA, IND, KOR, TWN, THA) at 3× costs plus 50 bps borrowing cost, short-only net SRs are negative for *every* strategy (MomSort −0.44, Ridge −0.76, XGBoost −0.98): the short leg's 19–58% turnover overwhelms any signal, reinforcing long-only as the relevant EM evaluation.

*Quarterly rebalancing.* Quarterly rebalancing lifts Ridge_All from 0.35 to 0.48 at 11% turnover, matching Ridge_Slow (0.47, 10%) with no feature selection at all; Ridge_Slow_Q reaches 0.49 at 7% and MomentumSort_Q barely changes (0.37). We do not discount this: it is what the frontier predicts (trading a third as often shrinks turnover regardless of signal mix), and it confirms the ceiling is mechanism-agnostic. The two routes are performance-equivalent in our sample but differ operationally: slow-feature selection retains monthly reactivity and an ex-ante diagnostic, while quarterly rebalancing holds stale fast-feature positions for three months and is validated only ex post. Stacking both adds little (0.49), consistent with substitutes on the same frontier.

*Statistical significance.* Jobson-Korkie tests with the Memmel [12] correction are significant at 5% in no individual market for MomSort > Ridge (aggregate Wilcoxon $p = 0.45$) or MomSort > XGBoost ($p = 0.008$), and in 1/15 markets for MomSort > LightGBM ($p = 0.003$); individual markets lack power at 55–93 OOS months. For Ridge_Slow vs. MomSort, we pool monthly net returns across all 15 markets and apply a block bootstrap (10,000 replications, resampling dates to preserve cross-market dependence): one-sided $p = 0.10$, 95% CI $[-0.08,\ 0.51]$ for the annualized SR difference. Ridge_Slow outperforms in 90% of bootstrap samples and 11/15 markets; a $t$-test on pooled monthly differences gives $t = 0.77$ ($p = 0.22$). The effect size ($d = 0.08$) would require ∼980 months of OOS data for 80% power at $\alpha = 0.05$: no feasible test can detect an improvement of this magnitude at conventional levels. The frontier claim is, however, jointly testable: sign-flipping the eight interventions' per-market SR differences vs. MomentumSort at the market level ($2^{15}$ exact patterns) gives mean +0.11, $p = 0.008$ (Wilcoxon 0.015). Since the eight variants span four mechanism families and markets are correlated, we read this as consistent joint evidence, not eight independent confirmations. Excluding the smallest universe (GRC, $N = 50$) leaves the hierarchy intact (Ridge_Slow 0.45 vs. MomSort 0.36; band 0.42–0.47), as does value-weighted construction (Ridge_Slow 0.40 vs. MomSort 0.31, Ridge_All 0.30; wins 10/15, positive 14/15).

*Feature classification sensitivity.* A data-driven persistence filter (rank AC > 0.75; ∼120 features) and a speed-tercile split (∼127 features) both yield net SR 0.47 at 13% turnover, matching the ad-hoc classification despite retaining 2× as many features.

## 7 Conclusion

We show that ML equity strategies in emerging markets face a cost-speed ceiling: across 15 markets and eight intervention variants, net Sharpe ratios converge on ∼0.45–0.49 at 3× EM transaction costs. The ceiling is traced by a linear frontier (Equation 2): at each fixed cost level, the ex-ante Feature Speed Index explains 64–65% of cross-sectional SR degradation ($\lambda = 0.33$ per unit $\kappa$; pooled $R^2 = 0.72$) and 73% of realized turnover. Non-smooth models (trees and unconstrained NNs) fall systematically below the frontier, confirming model smoothness as an independent cost channel.

The frontier implies that the problem is *not* which intervention to choose, but whether a strategy minimizes its Feature Speed Index. Feature selection, prediction smoothing, and reduced rebalancing frequency all reach the same ceiling through different mechanisms. Restricting Ridge to slow features is the simplest path (SR 0.47, 10% turnover); walk-forward EMA independently achieves 0.45.

Model complexity introduces a second constraint: tree models require output smoothing (EMA) to reach the ceiling, because piecewise-constant surfaces generate turnover independent of input stability.

More broadly, the ceiling validates Gârleanu and Pedersen's [7] prediction that optimal portfolios overweight slow signals, and quantifies the frontier that any practical ML strategy must respect.

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